Understanding Maturational Differences in Young Athletes

From R. Martens and R.S. Vealey, *Successful Coaching,* 5th ed. (Champaign, IL: Human Kinetics, 2024), 38-43.

Youth sport athletes are undergoing the most significant physical maturational changes that they will experience in their lives. Athletes' physical maturation does not proceed in unison with their chronological age. Chronological age is how old athletes are based on their dates of birth, whereas maturational age is how close they are to their full mature state. Athletes of the same chronological age can differ in maturation by as much as five years, based on their personal biological clocks. Two athletes may both be 12 years old (chronological age), but one may be 10 and the other 15 in terms of maturational age.

The range of individual differences in physical structure is greater between the ages of 10 and 16 than at any other time in the human life span. This is a period of very fast growth, although boys and girls differ in the timing of their growth spurts. The average age for the start of the growth spurt is 9.5 years for females and 11.5 years for males. Girls experience the fastest growth in height (or growth spurt) at an average age of 11 years, and boys at an average age of 13 years. As a result, it is common for girls to be taller than boys between ages 10 and 13 years old.

A common belief is that when adolescents go through their growth spurts, they experience awkwardness (lack of agility, balance, and coordination) until their bodies accommodate the changes that have occurred. Scientific evidence, however, has not supported this view. Some individual kids may decline on certain motor tasks during adolescence, and it may be that rapid growth in the feet can temporarily disrupt agility and coordination. However, the idea that young athletes need to relearn skills after their growth spurt is a myth; athletes quickly learn to adapt to their growing dimensions.

Assessing Maturation in Young Athletes

The most practical way to organize youth sports is by chronological age, and that's okay, as long as coaches are aware of maturational differences within groups. How should coaches assess the maturation of their athletes? There are several scientific methods, but the easiest way for coaches to do this is to simply observe their athletes.

Relatively taller athletes obviously have begun or completed their growth spurts. Typically, the height spurt (average of ages 11 for girls and 13 for boys) occurs first, and then is followed by weight gain and body changes that accompany puberty.

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Understanding Teenagers

Coaches, like parents and guardians, shake their heads at times in observing teenagers' moodiness, impulsiveness, focus on short-term gratification, and risk-taking. These behaviors are the result of the differences in the teen brain from the adult brain.

The brain's prefrontal cortex is not fully developed in teenagers and typically doesn't fully mature until the mid-20s (Casey & Caudle, 2013). This part of the brain helps to inhibit impulses, to understand the consequences of action, and to plan and organize behavior. While most adults use rational processes when facing emotional decisions, teenagers are not as biologically equipped to think through things in the same way. Teen brains also have highly active reward centers, which make teens more interested in seeking short-term rewards and taking risks.

Realizing that teenage athletes' impulsive and emotional responses are related to biological maturation of their brains may help coaches cope. Coaches could help athletes identify strategies to manage negative emotions that occur in competition, such as taking a deep breath and then exhaling the tension and anger. Athletes could also repeat a go-to word or phrase to lock their focus externally back on the task (e.g., "next play"), instead of being internally distracted by their feelings. Coaches can teach athletes to calm and recenter themselves through breathing and staying present-moment focused on the game when athletes come off the ice, floor, or field for a break or between shifts. Coaches can be supportive guides and offer encouragement for their athletes during these teenage years.

The body changes that coaches can observe are secondary sex characteristics, or the physical features that emerge during this transitional period between childhood and adulthood.

Example secondary sex characteristics in females are breast development and increased weight, while males have increased weight and muscle mass, facial and body hair, deepening of voice, and broader shoulder girth. Of course, such markers as pubic hair for boys and girls, as well as time of menarche (first menstrual period) are characteristics of maturation, but these are too invasive for coaches to use as indicators.

Early- and Late-Maturing Athletes

Young athletes can be categorized into three maturity groups—average, early, and late—according to how quickly they are growing and reaching maturation. Most kids fall into the average maturity category, meaning their maturational age is within one year of their chronological age. Early maturity is indicative of those whose maturational ages are at least one year greater than their chronological ages. Late maturity comprises those whose maturational ages are at least one year less than their chronological ages.

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Coaches should pay particular attention to early- and late-maturing athletes; these athletes experience both positive and negative consequences as a result of their maturational patterns. The following descriptions are general in nature based on typical maturational patterns, although not all adolescents neatly fit into these categories.

Early-Maturing Boys

Early-maturing boys tend to be taller and heavier, with more muscle mass and shoulder girth, all of which give them advantages in strength and power activities. These boys are often able to outperform their peers in many youth sport activities. Due to this early success, early-maturing boys receive recognition, which fuels their confidence and motivation. However, later-maturing boys often catch up to early-maturing boys in biological maturation and thus performance. It is sometimes difficult for the early-maturing boys to understand that their early biological advantage is gone.

Generally, early-maturing boys end up as shorter adults as compared to late-maturing boys, and early-maturing boys also average greater body weight and greater weight-for-height as adults than other maturity groups.

Late-Maturing Boys

Late-maturing boys, who tend to be smaller in stature and build than early-maturing boys, may not be as successful in early team-oriented youth sport experiences due to their relative lack of strength and power. They may be successful in sports such as cross country and diving in which smaller bodies are an advantage. Late-maturing boys are often teased, and their relative lack of success in sport may discourage them from continued participation. Ironically, late-maturing boys (and girls) tend to be taller as adults because they grow for a longer period of time.

Early- and Late-Maturing Girls

Like boys, early-maturing girls tend to be bigger, while late-maturing girls are smaller and thinner. Early-maturing girls will have greater strength and power, although maturation can also add fat weight that can be problematic in sports like gymnastics, diving, and figure skating. Early-maturing girls have been shown to have lower self-esteem and are more dissatisfied with their body image compared to later-maturing girls (Negriff & Susman, 2011). Similar to boys, late-maturing girls have a longer period of steady leg growth than early-maturing girls, and generally are taller adults. Early-maturing girls tend to be shorter adults, and early maturation is related to overweight and obesity in females.

How Maturational Level Affects Sport Participation

Maturational level influences youth athletes' sport participation, providing various advantages and disadvantages.

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Maturational Advantages in Specific Sports

The maturity levels of young athletes often predict who plays which types of sports as well as positions within sports. Youth baseball players who participated in the Little League World Series and on elite teams had older maturational ages and thus were bigger and taller compared to the average of their age groups (French et al., 2007). In particular, pitchers, shortstops, and first basemen were maturationally advanced with larger body sizes than athletes who played other positions. The same results were found with junior rugby players, who were taller, heavier, and more mature than the normal population (Till et al., 2010). Players' maturation also differed by position, with the forwards being bigger and more advanced in maturation than other positions.

An opposite maturational trend occurs in sports like figure skating and gymnastics, where athletes are judged on aesthetic appeal and late maturation provides an advantage (for girls) in terms of smallness and physical linearity. Eighty percent of junior elite figure skating participants were found to be average or late maturing (Monsma, 2008). Overall, the maturational level of athletes often influences the types of sports they engage in, the positions they are assigned to, and the levels of early success experienced.

The Relative Age Effect

It's common and practical in youth sport to group athletes into one-year (10-year-olds) and sometimes two-year (11- to 12-year-olds) age groups for teams and leagues. The intent in age groupings is to provide developmentally appropriate instruction and fair competition. However, differences in age between athletes born in the same calendar year, termed *relative age,* have been shown to have a major effect on the success that young athletes achieve in sport.

The relative age effect (RAE) occurs when athletes born early in an eligibility year (relatively older athletes) are overrepresented on sport teams as compared to those athletes born later in the eligibility year. This means that the most players in a program or league were born in the first few months of the eligibility year (e.g., January to April for ice hockey). The RAE was first noted in elite ice hockey, including the National Hockey League, and has continued to occur in both men's and women's professional as well as youth ice hockey (Hancock, 2017; Nolan & Howell, 2010). The RAE is a worldwide phenomenon shown in several sports, including soccer, basketball, cricket, rugby, and baseball (Vealey & Chase, 2016).

Why does the RAE occur? It begins with the advantage created by physical and mental maturation. Older athletes are bigger, stronger, and faster, and have more experience in skills in the sport and playing the game. These relatively older athletes experience more success, and thus receive more attention and more playing time, often in key positions. This success qualifies them for more select teams with stronger competition and better instruction.

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Why is the RAE a problem? Like late-maturing athletes, relatively younger athletes are prime candidates to become discouraged and drop out of sport. However, some of these athletes could develop outstanding skills and talent in their sports. We are overlooking a significant number of young athletes simply because we subconsciously favor those who are maturationally advanced. Ironically, late-maturing athletes tend to be taller adults, and could develop into an ideal physical size for their sports.

The RAE is difficult to overcome, but what is important is that coaches understand it and help each athlete develop their unique abilities regardless of skill or maturational level. This is what Athletes First, Winning Second is all about, as opposed to playing only the better, relatively older athletes to ensure winning. Committing to developmentally appropriate practice means understanding the RAE and other maturational influences on kids' participation in sports.

Bio-Banding

U.S. Soccer and the English Premier League instituted bio-banding in their development academies to enhance individual player development and help control for major differences in maturation within teams. In bio-banding, players are grouped (banded) based on their individual maturity and biological age rather than chronological age alone. Players' maturational levels are determined using a formula that calculates the current percentage of height of each player in relation to their predicted final adulthood height. Thus, one player may be at 84.5 percent of their predicted adult height, while another player may be at 98 percent.

The intent of bio-banding is to allow late-maturing athletes to play without being physically dominated, so they can express themselves creatively, technically, and tactically. Bio-banding also is targeted to enhance the development of early-maturing athletes who often over-rely on their physical advantage and neglect to develop the technical and tactical skills that will be required when their physical advantages no longer exist.

How has it worked? Here are some results from bio-banded soccer games:

- Early-maturing players said they had to place more emphasis on tactics and techniques, while late-maturing players felt they could better develop their technical, tactical, and physical skills (Cumming et al., 2018).
- Early-maturing players made more short passes and dribbled significantly less, while late-maturing players made more tackles and fewer long passes (Abbott et al., 2019).
- Coaches noted that bio-banding resulted in a game that was more technical, tactical, and team-oriented, and that they were able to view the skills of late-maturing players who typically were dominated by early-maturing players (Cumming et al., 2018).

These results illustrate how coaches can improve athletes' skill development by taking their maturational levels into account. Coaches don't necessarily need to bio-band athletes on their teams, but they should be able to observe how physical maturity influences athletes'

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Bio-Banding *(continued)*

performance. The key is to focus on athletes' technical and tactical skills—not their raw, physical strength and power, or lack of it.

Maturational Coaching Recommendations

1. Train athletes' skills, not their brawn. Understand the difference between physicality and skill and provide feedback on athletes' performance in relation to their skill, not the outcomes they achieve due to their physical size.
2. Encourage late-maturing athletes by reminding them that they will catch up and that their biological clock is just a little slower than others. By focusing on their skill development now, their performance potential will be enhanced when they do achieve physical maturity. Help them to focus on individualized performance goals in relation to their own improvement.
3. Because early-maturing athletes are bigger and stronger than their peers, they may fail to learn fundamental techniques that are needed for more advanced skill development later in their athletic careers. Coaches should emphasize personal mastery and individualized performance goals and not allow them to rely on their physicality and short-term biological advantage.
4. Avoid pigeon-holing prepubescent athletes into certain sport positions, because their early maturational heights and builds may not fit the position they were forced into as a youngster. It would be detrimental for youth basketball coaches to designate early-maturing boys as post players and not teach them perimeter shooting skills. Similarly, young athletes should not be pigeon-holed into a specific sport based on their body types. Youth athletes should sample multiple sports and activities at a young age, and avoid early sport specialization.

   Professional basketball superstar Anthony Davis grew eight inches his junior year of high school and changed from a 6-foot-2-inch (188 cm) perimeter guard to a 6-foot-10-inch (208 cm) shot blocker and rebounder. Because he had developed his outside shooting and ball handling skills as a guard, he was a complete player and number one NBA draft choice. Had he grown earlier, chances are a coach may have placed him exclusively in a post position, which may have enabled the team to win more games, but would have hampered Davis's skill development.

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Which maturational and cultural differences do you need to learn more about to understand your athletes?

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5. Counsel early-maturing girls whose body development typically is accompanied by increased sexualization. Sexualization is when girls receive societal messages that their worth and value is dependent upon their physical attractiveness, which leads to lower self-esteem, poor body image, and even eating disorders. Encourage early-maturing girls to stay involved in sport and teach them how to stand up to derogatory comments and harmful cultural messages.

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